COORDINATION

SUBJECT: CHEMISTRY **DATE: 05-04-2024**

1. Which of the following statement(s) is/are correct?

(A) $[Cr(NH_3)_6]^{3+} > [Mn(CN)_6]^{3-} > [V(CO)_6]$, With respect to magnetic moment (spin values in B.M.)

(B) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$, With respect to Δ_0 values.

(C) $[Ni(CO)_4] > [Co(CO)_4]^{-} > [Fe(CO)_4]^{2-}$, With respect to strength of M – C, π-bond. (M = Ni, Co or Fe)

(D) $[NiCl_4]^{2-} < [CuCl_4]^{2-} < [ZnCl_4]^{2-}$, With respect to stability.

Paragraph for (Ques. no. 2 to 4)

In coordination chemistry there are a variety of methods applied to find out the structure of complexes. One method involves treating the complex with known reagents and from the nature of reaction, the formula of the complex can be predicted. An isomer of the complex $CO(en)_2(H_2O)Cl_2Br$, on reaction with concentrated H_2SO_4 (dehydrating agent) it suffers loss in weight and on reaction with $AgNO_3$ solution it gives a white precipitate which is soluble in NH_3(aq).

2. The correct formula of the complex is:

(A) $[CoClBr(en)_2]\ H_2O$

(C) $[CoBr(en)_2(H_2O)]\ Cl_2$

(B) $[CoCl(en)_2(H_2O)]\ BrCl$

(D) $[CoBrCl(en)_2]\ Cl.H_2O$

3. If all the ligands in the coordination sphere of the above complex be replaced by F^-, then the magnetic moment of the complex ion (due to spin only) will be-

(A) 2.8 BM (B) 5.9 BM (C) 4.9 BM (D) 1.73 BM

4. If one mole of original complex is treated with excess $Pb(NO_3)_2$ solution, then the number of moles of white precipitate (of $PbCl_2$) formed will be-

(A) 0.5 (B) 1.0 (C) 0.0 (D) 3.0

5. Geometrical shapes of the complexes (p), (q) and (r) formed by the following reactions respectively are:

$$Cu^{2+}(aq) + KCN(aq)\ (excess) \longrightarrow \text{complex (p)}$$

$$CO^{2+}(aq) + KNO_2(s) + H^+(aq) \longrightarrow \text{complex (q)}$$

$$Zn^{2+}(aq) + NaOH(aq)(excess) \longrightarrow \text{complex (r)}$$

(A) Tetrahedral, octahedral and square planar
(B) Tetrahedral, octahedral and tetrahedral
(C) Square planar, octahedral and tetrahedral
(D) Octahedral, octahedral and tetrahedral

6. In which of the following pair(s) both the complexes show optical isomerism ?

(A) cis-$[Cr(C_2O_4)_2Cl_2]^{3-}$, mer-$[Co(gly)_3]$

(B) $[Co(en)_3]Cl_3$, cis-$[Co(en)_2Cl_2]Cl$

(C) $[PtCl(dien)]Cl$, $[NiCl_2(PPh_3)_2]$

(D) $[Co(NO_3)_3(NH_3)_3]$, cis-$[Pt(en)_2Cl_2]Cl_2$

7. Number of chelate rings in $[Fe(edta)]^- = a$

Number of chelate rings in $[Co(en)_2(NH_3)Cl]^{2+} = b$

Number of chelate rings in $[Ni(dmg)_2] = c$

Number of chelate rings in brown ring complex $[Fe(H_2O)_5NO_2SO_4] = d$

Calculate the value of (a + b + c + d) is .....

8. The volume (in mL) of 1.0 M $AgNO_3$ required for complete precipitation of chloride ions present in 30 mL of 0.01 M solution of $[Cr(H_2O)_6]Cl_3$, as silver chloride is close to-

ANSWER KEY											
Ques.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**			
Ans.	**ABD**	**D**	**C**	**A**	**B**	**AB**	**9**	**9**			